

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

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Abstract

001 Segmentation models in medical imaging are trained on one
002 acquisition setting and must generalize to others. Label-
003 generative domain randomization, of which SynthSeg is the
004 archetype, addresses this by discarding the acquired scan
005 and resampling intensities from a per-label Gaussian mixture
006 over a dense anatomical label map – an annotation far
007 richer than the segmentation task itself provides, and one
008 that does not preserve texture: real gradients and partial-
009 volume detail become piecewise noise. We ask when that
010 loss costs accuracy, and find the answer is a property of the
011 task. Where the target is separated from its surroundings
012 by an anatomical interface – encapsulated abdominal or-
013 gans, bounded by fat planes – destroying texture costs nothing.
014 Where no interface exists – diffuse glioma, infiltrating
015 beyond both the enhancing margin and the peritumoral
016 FLAIR abnormality – the target must be inferred from tissue
017 appearance, and destroying texture is expensive. We
018 exploit this with PALETTE, a single-source augmentation
019 needing no dense label map: it partitions a scan’s fore-
020 ground by unsupervised k -means, sub-divides each class
021 spatially by a Voronoi partition, and applies a signed random
022 affine remap to the real intensities of each sub-region.
023 Remapping existing intensities rather than replacing them
024 leaves edges, gradients and partial-volume detail largely
025 intact; only the contrast relationships between regions are
026 randomized. A controlled ablation isolates this: holding
027 the partition and per-region statistics fixed and swapping
028 only the real-intensity remap for label-style noise fill
029 costs 7.7 and 7.2 Dice on tasks with no tissue interface,
030 against +1.1 and -1.1 on organ tasks with one, and -4.5
031 and -1.4 on a third, far more densely-labelled interface
032 task (31-region healthy-brain parcellation) whose regions
033 are, on inspection, tissue-interface targets too – negative
034 rather than merely small, plausibly because so much more
035 of the volume is relabelled at once, but never a significant
036 gain either way. Across cross-contrast, cross-modality
037 and cross-dataset transfer on six tasks — healthy-brain,
038 MS-lesion, brain-tumour, abdominal-organ, mandible and

breast-lesion segmentation — one nnU-Net configuration
trained with PALETTE attains the best Dice on four, and improves
over all five competing methods on a task-stratified test. On the
other two it places second: a statistical tie on mandible, and a real
if small loss on breast, where the texture intervention nonetheless
remains among the strongest in the study.

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1. Introduction

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Medical image segmentation models are trained on one acquisition
setting and are frequently applied to another: a different scanner,
sequence, or imaging modality than the one seen during training.
MRI contrast is a protocol choice rather than a physical constant,
so the same anatomy yields incompatible intensity distributions
across sequences, and a network that has only seen T1-weighted
brain images can fail outright on T2-weighted or FLAIR images
of the same anatomy [3, 34]. Acquiring and annotating training
data for every contrast a model might encounter is not practical:
manual segmentation is expensive, and clinical sites rarely
share raw data across institutions [33, 52]. The single-source
domain generalization setting — train on one labelled dataset
in one contrast, generalize to unseen contrasts and modalities
without any target-domain data — is therefore the setting that
matters for deployment, and the one a growing body of work
addresses [36, 46, 56, 58].

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The dominant single-source strategy is label-generative domain
randomization, popularized by SynthSeg [3]. Given a dense
anatomical label map — a voxel-wise parcellation of the whole
image, not the sparse structure-of-interest annotation a
segmentation dataset normally provides — intensities are
resampled from a per-label Gaussian mixture, producing images
of arbitrary, unrealistic contrast that force the network to
ignore contrast and rely on shape and layout [3]. This is
effective, but carries two structural costs. First, it is *label-driven*:
the dense parcellation it resamples from is a richer annotation
than the task itself requires, so generating synthetic contrast
costs exactly the expensive la-

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076 belling the method aims to reduce reliance on. Second, and
077 less discussed, it *does not preserve texture*: replacing every
078 labelled region with piecewise Gaussian noise discards the
079 real gradients, partial-volume boundaries, and unlabelled
080 fine structure the acquired scan contains. Layout survives;
081 texture does not.

082 We ask whether discarding texture is necessary for con-
083 trast robustness, or merely the path of least resistance once
084 a dense label map is assumed. Our central finding is that
085 the answer depends on the *task*: large gains from preserv-
086 ing real intensity structure occur only where the segmenta-
087 tion target is defined by tissue appearance rather than sep-
088 arated from its surroundings by a genuine anatomical in-
089 terface. That texture carries diagnostic signal in exactly
090 the tasks where no such interface exists is established in-
091 dependently of us — texture features discriminate MS le-
092 sions from normal-appearing white matter [57], and diffuse
093 glioma infiltrates beyond every margin conventional MRI
094 shows [17, 48] — and our contribution is to show that this
095 distinction governs where a texture-preserving augmenta-
096 tion can help substantially.

097 We exploit it with **PALETTE**, a single-source augmen-
098 tation that needs no dense label map. **PALETTE** parti-
099 tions the foreground of a scan into intensity classes with
100 unsupervised 1-D k -means, further subdivides each class
101 into spatial sub-regions with a Voronoi partition, and ap-
102 plies a signed random affine remap to the *real* intensities
103 within each sub-region. Because the transform remaps ex-
104 isting intensities rather than replacing them with sampled
105 noise, edges, gradients, and partial-volume detail largely
106 survive within each sub-region; only the region-to-region
107 contrast relationship is randomized, including sign inver-
108 sions no physical acquisition protocol produces. The op-
109 tional final step is the only one that uses labels, and it uses
110 only the sparse task labels already present in any segmenta-
111 tion dataset — never a dense parcellation.

112 We test the mechanism claim directly rather than infer-
113 ring it from aggregate scores. Holding the k -means/Voronoi
114 partition and the per-region statistics fixed, we replace
115 only the real-intensity affine remap with label-style noise
116 fill, matching SynthSeg’s resampling inside **PALETTE**’s
117 own geometry, across fourteen training settings. Every
118 large effect of that swap falls on a target without a tis-
119 sue interface — +13.3 and +7.2 Dice on brain tumour,
120 +7.7 on MS lesions — while no interface-bounded tar-
121 get gains more than 1.4. The converse does not hold:
122 several appearance-defined settings, breast lesions among
123 them, show no reliable effect, so we state the result in one
124 direction only (§4.3). An independent, segmentation-free
125 gradient-orientation measurement confirms that texture is
126 what changes at the fill-swap step: the noise-fill steps leave
127 it at the noise floor while Dice still rises, and only the real-
128 fill swap restores it.

Empirically, training a single nnU-Net [22] configura-
tion — fixed across datasets, with only the epoch bud-
get varying — we evaluate cross-contrast, cross-modality,
and cross-dataset transfer on eight segmentation tasks:
healthy brain, MS lesion, brain tumour, abdominal or-
gan, mandible, breast lesion, spine and pelvis. We com-
pare against a no-augmentation baseline, both variants of
SynthSeg (the original dense-label-driven variant, and an
EM variant with no public implementation that we re-
implement), SRCSM [49], and Auglab [34], a published
augmentation pipeline designed for cross-modality training
and the strongest competitor. **PALETTE** is the best method
overall: it attains the best Dice on six of the eight tasks and
is significantly better than every competitor across tasks, on
both Dice and boundary distance. No competitor is reliably
second: the strongest alternative switches between Auglab
and SRCSM from task to task, so **PALETTE** is the only
method at or near the top throughout.

In summary, we make the following contributions:

- **A task-level account of when texture preservation matters.** Large gains from preserving real intensity structure occur only where the segmentation target has no anatomical interface separating it from its surroundings. We test this with a controlled intervention rather than asserting it post hoc, and report where it does not extend.
- **A causal ablation isolating texture.** Holding the intensity and spatial partition and the per-region statistics fixed and swapping only the real-intensity remap for label-style noise fill, across fourteen training settings, confines every large effect to no-interface targets. An independent gradient-orientation measurement confirms that texture, not the partitioning scheme, is what changes at that step.
- **PALETTE**, the single-source contrast augmentation that exploits this, requiring no dense anatomical label map and using only the sparse task labels a segmentation dataset already provides.
- **Breadth evidence.** Across cross-contrast, cross-modality and cross-dataset transfer on eight tasks and sixteen datasets, with one configuration reused throughout, **PALETTE** is best on six tasks and significantly better than every competitor across tasks. Where it does not win we report whether the method ahead is actually ahead: on healthy brain it is a tie, on mandible a real loss.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabelled target images and solves an easier problem [45]. Most medical DG methods are *multi-source*, interpolating across several labelled

domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [2, 30, 33, 35, 43, 52] – all of which still rely on source diversity. We instead work in the *single-source* regime — one labelled dataset, no target data, no second source – the setting that matters when target-domain data is unavailable at training time, which is routine in clinical deployment [36, 58]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [22].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs: it is *label-driven* (a dense anatomical label map is required per subject – SIAM shifts the dependence from label *quantity* to *quality* with curated templates, but each still needs a full dense parcellation [50]), and it is *texture-destroying*, since per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter and MS lesions and to atlas-based expectation-maximization [6, 7, 28]. Among SynthSeg’s own variants we take the original (*noEM*), which resamples intensities from a dense anatomical label map and never looks at the acquired scan, and the *EM* variant, which instead fits its per-class intensity model to the real image by expectation-maximization and so needs only the sparse task labels a segmentation dataset already provides, not a dense parcellation. The EM variant has no released implementation; ours follows the mechanism described in the original paper [3], and a faithful re-implementation is itself a contribution.

2.3. Texture as a segmentation signal

That local texture, not only intensity level, carries information a segmentation model can use is established independently of us: second-order gray-level statistics discriminate image content beyond simple intensity [18], and quantitative texture features from tumour regions carry prognostic signal across CT, MRI and PET [27]. Closer to our setting, Kang et al. disentangle structure from texture in cross-domain image translation with a texture co-occurrence loss, targeting the same failure mode we do — naive translation distorts fine texture and degrades downstream segmentation [24]; a texture-fidelity loss has likewise been added to GAN-based lesion augmentation [15]. That some targets resist boundary-based methods for lack of a sharp edge,

and are better served by texture-based segmentation, has been argued directly for diffuse white-matter lesions [26]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to *which* segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [22, 34] is one such GPU-accelerated framework and is the strongest competitor in our experiments, and BigAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [58]. Our method inserts the PALETTE transform as one additional operator inside the Auglab pipeline, so the comparison against Auglab directly isolates PALETTE’s contribution. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [36] (the GIN-alone configuration we compare against is an ablation in the original paper of a two-view objective on 2-D slices, so our comparison is illustrative of the random-filter family rather than a reproduction of that paper’s proposed method); related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or perturb the Fourier amplitude spectrum instead of the image itself [8, 31, 40, 46, 56, 59]. Our operator instead partitions the foreground into intensity classes and Voronoi subregions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [49] (semantic-aware random convolution with source matching), which also localizes its perturbation region-wise in true 3-D and, like PALETTE, layers exactly one label-consuming component over an otherwise label-free step: at training time it draws a fresh untrained convolution per anatomical class from the ground-truth map (needing a Gaussian smoothing pass to control the boundary-contrast artifacts this introduces), and at test time applies a label-free histogram matching onto the source distribution. The difference from PALETTE is not label use versus none, but *how* that step transforms the signal: an untrained nonlinear convolution versus a deterministic affine remap that leaves relative intensity structure inside each region intact.

2.5. Cross-contrast evaluation protocol

We evaluate a model trained on a single real input modality on every held-out modality of the *same* segmentation task, including the training modality itself, across eight tasks spanning brain, abdominal, dental, breast, spine and pelvic anatomy in MRI, CT and CBCT. This single-sequence-in, every-sequence-out design is not unprecedented – Auglab’s own evaluation follows the identical protocol for spine segmentation [34] – and our contribution is breadth (eight tasks, not one) and a further axis: wherever an external cohort exists, it is pooled into its task as additional held-out contrasts from other sites, so the training dataset’s own test set is one point among several rather than the only held-out evidence. For mandible, which is acquired as CBCT only, that cross-dataset transfer is the entire held-out axis.

3. Method

We describe PALETTE (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE transform

Let x denote the intensity volume of a training scan, min-max normalised to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real OnHarmony (brain MRI) volume. The figure’s five panels (a)–(e) are deliberately lettered rather than numbered 1–5 so they are never confused with the four method steps below: panels are *intermediate diagnostic renderings*, one per step plus the input, not a one-to-one visual walkthrough of the steps themselves (panel (e), the final training input, is the only panel that shows the texture the transform actually produces; (b)–(d) show flat preview values, defined in the caption).

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, panel (b)). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast – CT, T1, T2, FLAIR, or otherwise. This step is occasionally skipped in favour of a single global remap (step 3 below, with $C = 1$); per-step probabilities are set once and never tuned per dataset, and are listed in Sec. 7.

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, panel

(c) shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions and receive independent contrast perturbations. Each class is usually, but not always, sub-split this way (Sec. 7).

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is occasionally Gaussian-blurred, mostly not (Sec. 7).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised, so the k -means/Voronoi partition need not respect a true anatomical boundary: when two adjacent tissues share similar intensity, step 3 can by chance assign them the same target contrast and erase that boundary rather than randomizing it. This step corrects for that: for each real anatomical label ℓ present, we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ ’s voxels, **on top of the step-3 output** rather than the original voxels (Fig. 1, panel (d)) – so it composes with, rather than replaces, the unsupervised remap, and ℓ ’s voxels still inherit the internal structure steps 1–3 imposed on them. Because label assignment is independent of the unsupervised partition, this reintroduces a full-strength contrast discontinuity at the true label border regardless of whether step 3 preserved it, sharpening the boundary cue available to the network – not the underlying anatomy. It consumes only the labels the segmentation task already requires, so it needs no additional annotation, and is optional: disabling it (`label_remap_prob=0`) degrades PALETTE gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalisation follows an optional second blur pass, giving the training input shown in panel (e) of Fig. 1. Per-step probabilities are set once, never tuned per dataset, and listed in full in Sec. 7.

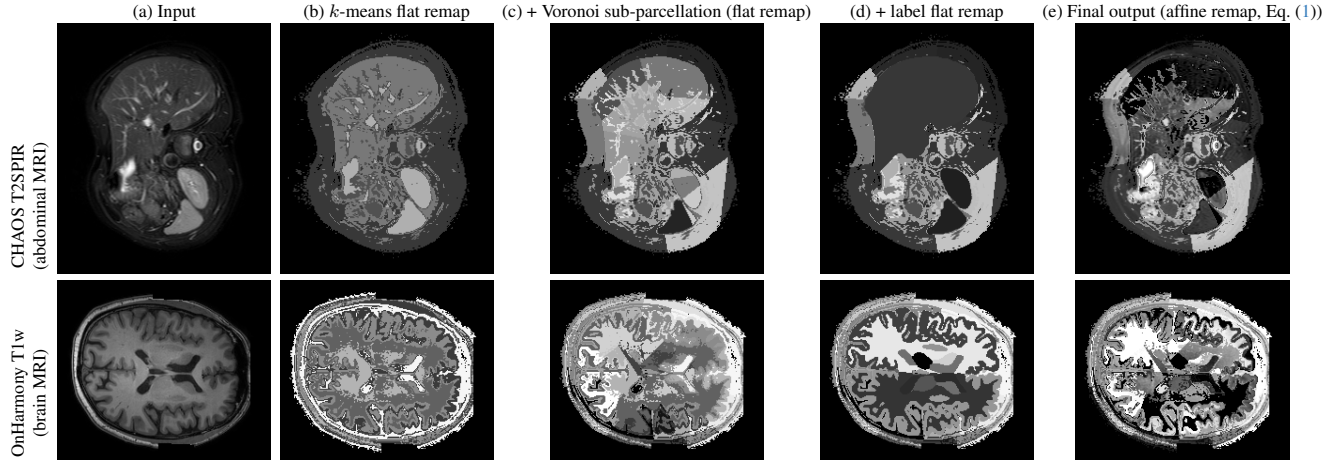


Figure 1. The PALETTE pipeline on one real CHAOS T2SPIR volume (top) and one real OnHarmony T1w volume (bottom); every panel is a deterministic function of the same real volume (no external model), min-max normalised to $[0, 1]$ for display only. Panels (b)–(d) paint each region, sub-region, or label with its precomputed target mean μ from Eq. (1), isolating that flat value from the texture applied on top of it; panel (e) is the actual training input, produced by applying the full signed affine remap of Eq. (1) with real texture (not a flat fill) at both the region and, where available, anatomical-label level.

3.2. Training protocol

PALETTE is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [22, 34], alongside that pipeline’s standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k -range, sub-region count, blur schedule, and α range given in §3.1 – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k -means + Voronoi vs. a supplied anatomical label map as the sole generative substrate) and the perturbation itself (an affine remap of real intensities vs. resampling from a per-label Gaussian mixture). Comparing PALETTE to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, that holds the k -means/Voronoi partition and the per-region statistics (μ_R, α_R) from Eq. (1) exactly fixed, but fills each sub-region with samples drawn from a Gaussian with that region’s $(\mu_R, \alpha_R \sigma_R)$ instead of remapping the real intensities inside it – i.e. the same partitioning scheme, with SynthSeg-style noise substituted for the affine remap. Because the partition and target statistics are identical to PALETTE’s own, any difference in downstream performance between PALETTE and noise-fill isolates the one

variable that changed: whether real texture is preserved or discarded within each region. We report this comparison separately for structures with no tissue interface (fine internal gradients, low boundary contrast) and structures bounded by one (identified by a sharp, high-contrast border) in Sec. 4, as the texture-preservation hypothesis predicts a widening gap specifically on the former.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the *same* segmentation task, using one fixed nnU-Net [22] configuration across all settings. No target-domain data, of any modality, is seen during training. Our full method, denoted **Ours** throughout, is PALETTE applied as one additional contrast-augmentation operator inside the Auglab pipeline [34]; the *Auglab* baseline is the same pipeline with the PALETTE operator removed, so the two differ by exactly the PALETTE transform. Every method is trained with the identical nnU-Net schedule — SGD with Nesterov momentum (0.99), the nnU-Net `polyLR` decay from an initial rate of 10^{-2} , 250 mini-batches per epoch, and the combined Dice-plus-cross-entropy loss — and differs only in the augmentation applied at the input. The epoch budget is fixed per dataset and shared by every method on that dataset (2500 for Brats-GLI, 2000 for ON-Harmony and Open-MS, 1000 for ToothFairy2 and I-SPY2, 200 for CHAOS and Pelvis; the Spine models were trained by our collaborators on their own pipeline with the same method suite, Sec. 8), and all results are 3-fold cross-validated, as

is standard practice in medical image segmentation and so that every comparison rests on per-case statistics rather than a single split. All significance tests are paired per case and Holm-corrected [20]; full test construction, configuration details, and per-dataset descriptions (subject counts, acquisition protocols, and label provenance) are given in Secs. 7 to 9.

Datasets. We evaluate on eight tasks drawn from sixteen datasets (Tab. 1): abdominal organs, healthy brain, brain tumour, MS lesions, mandible, breast lesions, spine and pelvic musculoskeletal anatomy, across MRI, CT and dental CBCT. Every task trains on a single source modality at a time and is tested on every other contrast of that task *and* the training contrast itself — eighteen source-modality settings in total; no target modality is seen during training. Wherever an external cohort exists, its contrasts are pooled into that task as further held-out test columns rather than reported as a task of their own: CHAOS [25] transfers to CT from AMOS [23] and SLIVER07 [19]; Mandible (ToothFairy2 [5], CBCT only) to the CT and MR of HaN-Seg [39] and the CT of PDDCA [41], which makes it the one task whose held-out axis is entirely cross-dataset; Breast (I-SPY2 [32]) to Duke-Breast-MRI [13, 42], I-SPY1 [21] and ACRIN-6698 [38], the last contributing diffusion-weighted MRI, a contrast neither training modality resembles; and Spine (a CT/Dixon cohort) to the T1w and T2w of SPIDER [51]. Pelvis is built from the TotalSegmentator CT and MRI releases [9, 54], two unpaired cohorts. ON-Harmony [53] reference labels are silver-standard, obtained on T1w with FreeSurfer [12] and registered to the other contrasts.

Baselines. SynthSeg comes in two variants, which we refer to throughout by how they obtain their per-class intensity model: *noEM*, the released variant, which reads it off a dense anatomical label map; and *EM*, which estimates it from the image by expectation-maximization and therefore needs no dense parcellation. We compare against five methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [3], the original label-generative variant (the implementation released by the authors), which synthesises each training image from a dense anatomical label map alone, ignoring the acquired scan entirely. (3) *SynthSeg-EM*, the variant described in the original work but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not dense-label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *SRCSM*, a semantic random-convolution contrast-synthesis augmentation, included as a second strong synthesis-based com-

Task	Dataset	Train modalities	Test modalities
Multi-Organ	CHAOS	T1-in, T2-SPiR	T1-in, T1-out, T2-SPiR, CT
	AMOS	—	CT, MRI
	SLIVER07	—	CT
Brain Glioma	Brats2024-GLI	T1n, T1c, T2w, FLAIR	T1n, T1c, T2w, FLAIR
Brain MS	Open-MS	FLAIR, T1w	FLAIR, T1w, T2w
Healthy Brain	ON-Harmony	T1w, T2w, DWI	T1w, T2w, BOLD, DWI, EPI, GRE
Mandible	ToothFairy2	CBCT	CBCT
	HaN-Seg	—	CT, MR-T1
	PDDCA	—	CT
Breast	I-SPY2	T1-CE, T2w	T1-CE, T2w
	Duke-Breast-MRI	—	T1-CE, pre-contrast
	I-SPY1	—	T1-CE, pre-contrast
	ACRIN-6698	—	DWI
Spine [§]	CT/Dixon cohort	CT, Dixon in	CT, Dixon fat/in/water
	SPIDER	—	T1w, T2w
Pelvis	TotalSegmentator	CT, MRI	CT, MRI

Table 1. Per-task dataset roster. Rows marked “—” are cross-dataset transfer targets only, never trained on; their contrasts are extra held-out test columns of the task they sit under, not tasks of their own. Mandible’s held-out axis is entirely cross-dataset, since ToothFairy2 acquires CBCT alone. Pelvis’s CT and MRI come from different patients. [§]Spine results were computed by our collaborators on their own pipeline, with the same method suite but without SRCSM and without HD95 (Sec. 8). Subject counts, acquisition protocols, and label provenance are in Sec. 8.

petitor that is neither label-generative nor a member of the Auglab family. (5) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [34]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE transform.

4.2. Main comparison

PALETTE is the best method overall. It attains the best Dice on six of the eight tasks, and across tasks it is significantly better than every competitor, including the strongest, Auglab ($p = 1.0 \times 10^{-13}$). Within individual tasks it significantly outperforms *every* competitor on four (Brats-GLI, Open-MS, Breast and Spine) and most of them on the remaining four (bottom row of Tab. 2). Every synthetic-contrast method vastly outperforms the no-augmentation baseline, confirming that cross-contrast augmentation is indispensable under single-source evaluation. SynthSeg-noEM, which resamples intensities from a *dense* anatomical label map, is competitive only where such maps are meaningful and collapses on the sparse-label tumour and lesion tasks, where no such map exists – a property of the method’s input requirements, not an implementation artifact. The re-implemented, label-using SynthSeg-EM closes most of that gap.

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	23.4	29.3	35.0	21.0	59.2	35.0	48.5	29.8	9.7e-75
SynthSeg-noEM	8.8	59.6	64.8	4.3	53.6	16.5	50.4	15.0	1.8e-85
SynthSeg-EM	43.1	83.0	66.3	33.9	76.3	39.6	80.6	62.5	1.4e-34
SRCSM	41.5	83.9	66.6	33.0	79.0	16.4	—	65.3	2.3e-86
Auglab	44.8	83.3	66.1	34.3	78.3	40.1	80.9	64.3	1.0e-13
PALETTE (ours)	45.9	85.0	66.5	38.5	77.7	41.2	82.3	65.6	—
competitors beaten ($p < 0.05$)	5/5	4/5	2/5	5/5	3/5	5/5	4/4	3/5	

Table 2. Task-level Dice (%). Each column pools that task’s held-out contrasts, all of its source modalities and, where one exists, its external cohorts, pooled by contrast group so that a modality tested by several external sources does not outvote one tested by fewer (Sec. 9). p : Holm-corrected one-sided sign-flip test on the macro-averaged per-patient difference, stratified by task with tasks weighted equally — a test *across* tasks, not an average *of* them; we deliberately print no cross-task mean Dice, since averaging a whole-liver Dice with a breast-lesion Dice yields a number no task’s clinician would recognise. The bottom row counts, per task, the competitors PALETTE beats with the same test run within that task alone. Spine has no SRCSM run (Tab. 1). HD95 is in Tab. 8.

No competitor is reliably second. The strongest competitor alternates between two methods with opposite failure modes: Auglab on Brats-GLI, Open-MS, Breast and Spine, and SRCSM on CHAOS, ON-Harmony, Mandible and Pelvis — yet SRCSM collapses on Breast (16.4 Dice). Choosing between them would require knowing the answer in advance. PALETTE is first or second on every task but one. We regard this breadth, rather than the size of any single margin, as the benchmark claim: a plug-in augmentation is useful precisely when it can be applied without knowing in advance which alternative would have been the right one for the task at hand.

Where PALETTE does not win, and by how much. The one-sided “ours is better” test cannot adjudicate a lead in the other direction, so on the two tasks PALETTE does not top we also ran the mirrored test, on the identical patient-level design. On ON-Harmony the result is a tie: SRCSM leads by 0.1 Dice and neither method is significantly ahead of the other ($p = 1.00$ both ways). On Mandible the loss is real: SRCSM (+1.3 Dice, $p = 6 \times 10^{-4}$) and Auglab (+0.6, $p = 2.5 \times 10^{-3}$) are both significantly ahead and PALETTE places third. The ablation of Sec. 4.3 locates that shortfall: on Mandible the texture step itself is significantly *positive*, and it is the subsequent addition of the Auglab recipe that costs Dice (Tab. 10). A benchmark ranking and a causal account are separate claims, and the second is the one this paper argues for.

4.3. Causal ablation: when does texture matter?

Boundary distance (secondary metric). HD95 is less standard than Dice for this class of comparison, so its task-level table is in the supplement (Tab. 8). In brief, PALETTE has the lowest HD95 on four of the seven tasks that report it (Brats-GLI, ON-Harmony, Open-MS, Pelvis) and is significantly better than every competitor across tasks,

Auglab included ($p = 5.1 \times 10^{-6}$). It ties Auglab on Breast ($p \geq 0.43$ both ways), trails SRCSM on CHAOS without the gap being significant, and on Mandible is significantly behind SRCSM, Auglab and the no-augmentation baseline — the same task, and the same cause, as its Dice shortfall.

PALETTE shares its k -means/Voronoi partitioning scheme with a much simpler procedure, so the central question is whether the partitioning alone — not the texture-preserving intensity remap — is what helps. We answer it with an intervention. Along a rung sequence in which each step adds exactly one ingredient (baseline $\rightarrow$ + k -means $\rightarrow$ +label-remap $\rightarrow$ +Voronoi with *noise* fill $\rightarrow$ same partition with *real*-intensity fill $\rightarrow$ +Auglab), every rung executes the identical pipeline of Sec. 3 in the identical order — the ladder only toggles which steps are active (e.g. the label-remap probability and the Voronoi sub-parcellation probability go from 0 to their default value one at a time), never reorders them. The fourth-to-fifth step changes *only the fill*: the partition, the number of sub-regions and the per-region statistics are held fixed, and label-style Gaussian noise is swapped for the real-intensity remap. The two Gaussian-blur passes (Sec. 3) are drawn from the same σ distribution at the same two points in the pipeline on both sides of this step, so blur is not confounded with the fill swap either. Any change at that step is therefore attributable to texture preservation alone.

Fig. 2 and Tab. 3 support a one-directional result. Every large gain from preserving real texture is on a target defined by tissue appearance rather than by an anatomical interface: Brats-GLI T1c (+13.3) and T1n (+7.2), both significant after correction over all fourteen rows, and Open-MS FLAIR (+7.7), positive for every one of its eight test patients. No tissue-interface row gains more than 1.4 Dice; the two that gain significantly, CHAOS T1in and ToothFairy2, do so by about a point. The converse does not hold, and we do not claim it: several appearance-defined settings show no reli-

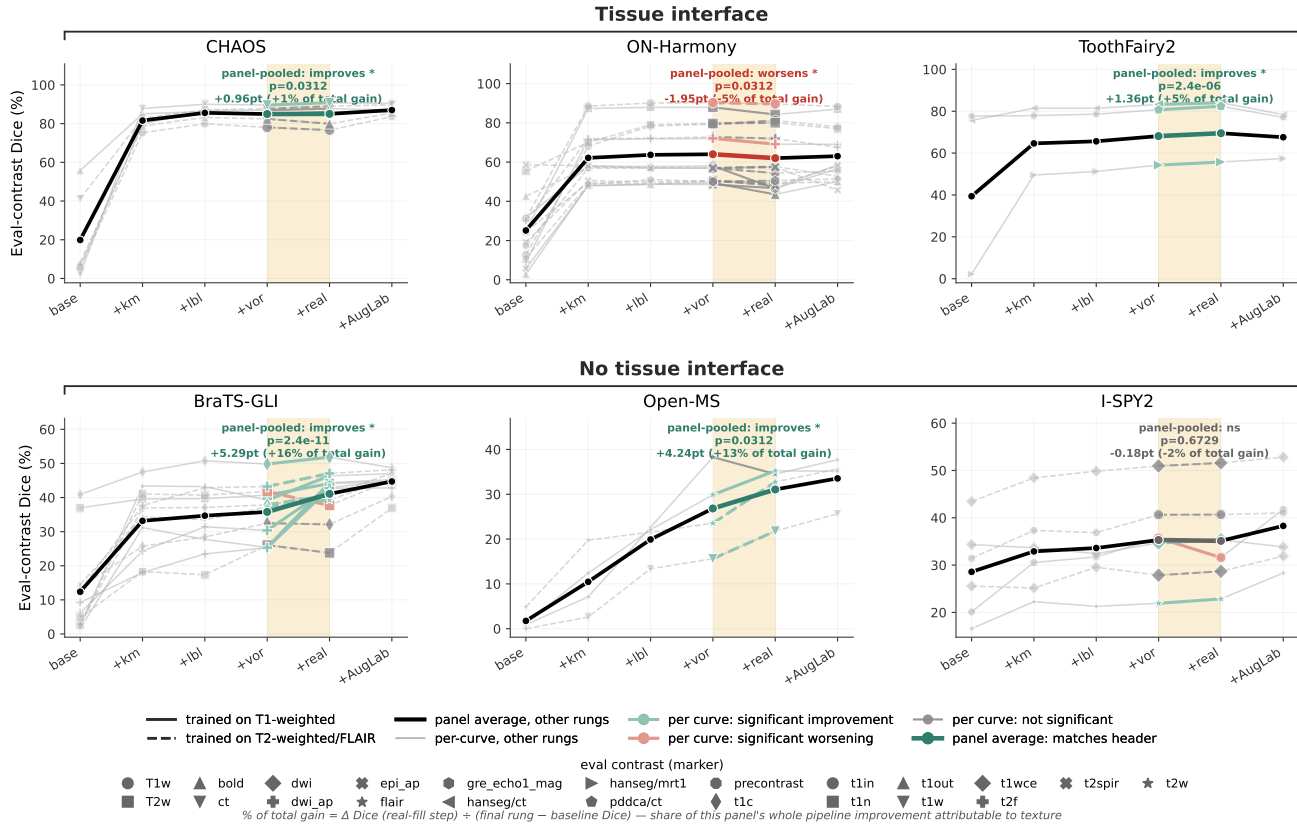


Figure 2. The one-variable test, one panel per task with a ladder, grouped by boundary type. Each thin line is one (training modality, held-out contrast) pair, external cohorts included, and only its real-fill segment (shaded band) is coloured, by that pair’s own significance at this one step – teal = significant improvement, red = significant worsening, grey = not significant. The bold black line is the panel average; its real-fill segment is coloured to match the panel-level test annotated above it, which pools all of the panel’s training modalities into one paired Wilcoxon test on *patients* (each counted once across contrasts and models), Holm-corrected across the six panels. Two of the three no-tissue-interface panels improve significantly (Brats-GLI $p = 2.4 \times 10^{-11}$, Open-MS $p = 0.031$) and Breast does not move. Among the tissue-interface panels, CHAOS and ToothFairy2 show small significant gains and ON-Harmony a significant worsening. Solid/dashed = trained on a T1-weighted vs. other contrast; marker = held-out contrast. Per-row numbers in Tab. 3; HD95 in Sec. 9.

able effect — Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows, where pooling five cohorts by contrast leaves the step at -0.85 and $+0.50$. A large texture gain is therefore diagnostic of an appearance-defined target, but an appearance-defined target does not guarantee one. This identifies *when* texture preservation can matter, and does so by intervention rather than by correlation across methods.

Does ON-Harmony’s negative effect reflect the same tissue-interface pattern as CHAOS, or a different phenomenon driven by how densely it labels the brain? ON-Harmony’s 31 regions – the standard subcortical parcellation (lateral, 3rd and 4th ventricles, cortex, white matter, cerebellar white matter and cortex, brainstem, eight deep grey nuclei) – are, without exception, bounded by a genuine tissue-type transition: CSF against tissue for the ventricles, grey against white matter for cortex and nuclei. There is

no texture-only label in the set, unlike a tumour or lesion margin, so ON-Harmony groups with CHAOS as a tissue-interface task, one that labels near-total brain volume rather than a handful of organs. Whether that labelling *density* has its own effect is a question one densely-labelled task cannot answer.

Does the fill swap change texture? Noise versus real fill.

The ladder shows that the fill swap changes Dice; it does not by itself show that what changed is texture. We measure that directly, without reference to any segmentation, using the normalized-gradient-field (NGF) similarity [16] between each augmented volume and its source scan — an alignment-of-gradients measure that is high when local edge and texture structure is retained and falls towards the isotropic-noise floor ($\approx 1/3$) when it is destroyed. Fig. 3

Task	boundary type	Δ Dice	p
CHAOS T1in	interface	+1.07	0.0014
CHAOS T2spir	interface	-0.88	0.17
ON-Harmony T1w	interface [†]	-5.18	0.078*
ON-Harmony T2w	interface [†]	-1.55	0.078*
ON-Harmony DWI	interface [†]	+0.78	0.11*
ToothFairy2 CBCT	interface [‡]	+1.36	5.8e-6
Open-MS FLAIR	no interface	+7.69	0.078*
Open-MS T1w	no interface	+0.79	1.0*
Brats-GLI T1n	no interface	+7.22	3.3e-11
Brats-GLI T1c	no interface	+13.27	5.0e-12
Brats-GLI T2w	no interface	+0.87	0.38
Brats-GLI FLAIR	no interface	-0.19	1.0
I-SPY2 T1-CE	no interface	-0.85	0.26
I-SPY2 T2w	no interface	+0.50	0.11

Table 3. Replacing label-style noise fill with the real-intensity remap, *holding the partition and per-region statistics fixed*: change in out-of-domain Dice for every training modality of every task with a ladder. p : paired Wilcoxon on patients, pooled over that row’s held-out contrasts, Holm-corrected across all fourteen rows. Every large gain is on a no-tissue-interface target; no interface row gains more than 1.4 Dice. *Open-MS has eight test patients and ON-Harmony eight test sessions (four subjects, each scanned at two sites), so no row of theirs can reach $p < 0.05$ after correction even when all eight move the same way, as they do for Open-MS FLAIR (8/8 improve) and ON-Harmony T1w and T2w (8/8 worsen). For ON-Harmony we treat the session as the unit; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic. [†]31 regions, each bounded by a genuine tissue-type transition (see text). [‡]Held-out axis is cross-dataset (HaN-Seg, PDDCA), since ToothFairy2 is CBCT only. HD95 in Sec. 9.

follows both quantities along the same rungs on Open-MS FLAIR. The three noise-fill rungs sit flat at the texture floor (NGF 0.45–0.46) while their Dice still climbs from 11.2 to 19.6; those gains come from richer contrast randomization, not from texture. Only the real-fill swap moves both together, NGF to 0.81 and Dice to 27.3. Table 4 places the methods on the same scale: PALETTE retains significantly more source texture than Auglab (FLAIR $p = 3.2 \times 10^{-3}$, T1w $p = 1.2 \times 10^{-6}$), while the label-style methods sit near the floor.

Why the margin over Auglab is the smallest. The same measurement explains why Auglab is the closest competitor: it is *not* a texture-destroying method, retaining NGF only modestly below PALETTE’s and far above the label-style methods’ (Tab. 4). Two methods that both largely preserve texture should land close together where texture is the operative cue, and they do; the large margins appear against the methods that discard texture.

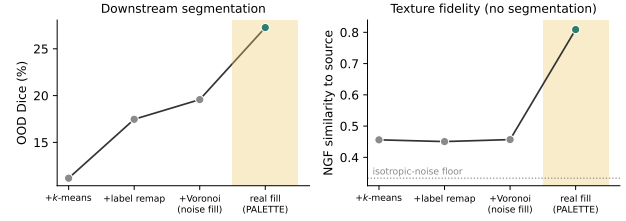


Figure 3. Noise versus real fill on Open-MS FLAIR. Left: out-of-domain Dice at each rung. Right: NGF similarity of the augmented volume to its source (mean $\pm$ s.e. over 30 scans), a segmentation-free measure of retained texture. The noise-fill rungs stay at the texture floor while Dice rises; the real-fill swap (shaded) lifts both.

Method	FLAIR	T1w
PALETTE	0.809	0.894
Auglab	0.768	0.777
SynthSeg-EM	0.462	0.422
noise fill	0.457	0.401
SynthSeg-noEM	0.330	0.319

Table 4. Texture fidelity: NGF similarity to the source scan (mean over $n=30$ Open-MS scans; $\approx 1/3$ = isotropic-noise floor). PALETTE retains the most texture; the label-style methods sit near the floor.

Two properties of the test. First, it compares two versions of the same method that differ in one component, so it cannot be explained by differences in augmentation strength, partition granularity or training schedule. Second, it is independent of whether PALETTE wins the task. Mandible makes that concrete: the texture step is significantly positive there (+1.36), yet the full recipe places third, because the following rung — adding the Auglab pipeline on top — costs Dice ($69.5 \rightarrow 67.6$, Tab. 10).

Supporting analyses. A component ablation separating the k -means partition, the Voronoi sub-parcellation and the signed remap is in Sec. 11. Per-setting tables for all eighteen source-modality settings, every rung of every ladder and the HD95 versions of Tab. 3 are in Sec. 9.

5. Discussion and Limitations

The causal ablation isolates a mechanism, not a full explanation. The noise-fill comparison in Sec. 4.3 shows that texture preservation, specifically, drives the advantage on no-tissue-interface structures; it does not by itself explain the smaller, more uniform gains observed on tissue-interface structures, which likely stem from the wider space of region-to-region contrast relationships (including the sign inversions in Eq. (1)) that no single real acquisition

660 protocol supplies. Disentangling these two effects further is
661 left to future work.

662 **Why the effect follows the anatomy, in one direction.**
663 Large gains from preserving real intensity structure appear
664 only where the target is defined by tissue *appearance* rather
665 than by an anatomical *interface*. Liver, spleen and kidney
666 are encapsulated organs bounded by fat planes – a gen-
667 uine physical interface, which is why edge-following meth-
668 ods such as geodesic active contours have long succeeded
669 on them [25, 47]; an augmentation that scrambles inter-
670 nal texture while preserving that interface loses little. Dif-
671 fuse glioma has no such interface: tumour cells infiltrate
672 beyond both the contrast-enhancing margin and the peritu-
673 moral FLAIR abnormality [17, 48], so the target must be in-
674 ferred from appearance rather than followed along an edge.
675 MS lesions are likewise defined by tissue signal charac-
676 teristics rather than an anatomical border [57]. This account is
677 an interpretation of the ablation, not an independent proof
678 of it; the causal evidence is the intervention in Sec. 4.3. It
679 also extends to the ON-Harmony result once its actual labels
680 are examined (Sec. 4.3): its 31 regions – ventricles, cor-
681 tex, white matter, deep grey nuclei – are bounded by gen-
682 uine tissue-type transitions throughout, so despite first ap-
683 pearances it groups with the organ tasks rather than the no-
684 interface ones. The larger negative magnitude there, com-
685 pared to CHAOS’s near-null one, is plausibly a density ef-
686 fect – near-total brain volume relabelled at once, rather than
687 a handful of organs – which we leave to a dedicated follow-
688 up with a second densely-labelled task.

689 **Appearance-defined is necessary, not sufficient.** Sev-
690 eral appearance-defined settings show no reliable texture ef-
691 fect: Open-MS T1w, Brats-GLI T2w and FLAIR, and both
692 Breast rows. We do not have a tested explanation, and we
693 state the result accordingly as one-directional. Two candi-
694 date factors are worth naming as hypotheses only. Within
695 one task the effect differs sharply by training modality
696 (Brats-GLI: +13.3 from T1c, +7.2 from T1n, about zero
697 from T2w and FLAIR), so how much discriminative tex-
698 ture the *training* contrast carries may matter as much as the
699 target. And Breast pools five cohorts from different trials
700 and sites, which may dilute a cohort-specific effect. Either
701 would be a property of the data rather than of the mecha-
702 nism, but neither is established here.

703 **The label step needs only the annotation training al-**
704 **ready assumes.** PALETTE’s one label-consuming step
705 uses whatever segmentation labels the training set happens
706 to carry – sparse, partial, or task-specific – and never a
707 dense anatomical parcellation. This is the practical dif-
708 ference from label-generative synthesis: SynthSeg’s dense
709 requirement means the augmentation costs an annotation

richer than the task, whereas here the step is free at train-
ing time and can be switched off entirely on an unlabelled
dataset, in which case PALETTE degrades gracefully to its
unsupervised steps.

Scope. PALETTE targets the single-source setting: one
labelled training modality, no access to target-domain data
of any kind. It is input-level and architecture-agnostic, and
we do not claim it subsumes multi-source domain general-
ization methods that have access to several labelled source
domains; the two are complementary rather than competing
solutions to the same problem.

6. Conclusion

We asked when preserving real image texture matters for
contrast-robust segmentation, and found that the answer is a
property of the task rather than of the augmentation: it mat-
ters where the target is defined by tissue appearance, and
not where it is defined by an anatomical interface. A con-
trolled intervention – holding the partition and per-region
statistics fixed and swapping only label-style noise fill for
a real-intensity remap – confines every large effect to no-
tissue-interface targets — +13.3 and +7.2 Dice on brain
tumour, +7.7 on MS lesions — while no interface-bounded
target gains more than 1.4; it does not, conversely, guaran-
tee a gain on every appearance-defined target. PALETTE,
the single-source augmentation built on this observation,
partitions the foreground by unsupervised k -means and a
Voronoi sub-parcellation and applies a signed affine remap
of the real intensities within each sub-region, requiring nei-
ther a dense anatomical label map nor any target-domain
data. Across eighteen source-modality settings spanning
eight tasks and sixteen datasets, a single nnU-Net config-
uration trained with PALETTE is the best method overall:
best Dice on six of the eight tasks and significantly better
than every competitor across tasks. It does not win every-
where: on healthy brain it ties SRCSM, and on mandible
SRCSM and Auglab are genuinely ahead, a shortfall the
ablation traces to the Auglab recipe PALETTE runs inside
rather than to the texture-preserving transform. A bench-
mark ranking and a causal account are separate claims; we
regard the task-level account as the more durable contribu-
tion: it predicts in advance where a texture-preserving aug-
mentation will and will not pay off, and that prediction is
testable on any new task.

References

- [1] Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A. Landman, Geert Litjens, Bjoern Menze, Olaf Ronneberger, Ronald M. Summers, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022. 3

- [2] Jie Bao, Zhixin Zhou, Wen Jung Li, and Rui Luo. Structure-aware stylized image synthesis for robust medical image segmentation. *arXiv preprint arXiv:2412.04296*, 2024. 3
- [3] Benjamin Billot, Douglas N. Greve, Oula Puonti, Axel Thielscher, Koen Van Leemput, Bruce Fischl, Adrian V. Dalca, and Juan Eugenio Iglesias. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. *Medical Image Analysis*, 86:102789, 2023. 1, 3, 6
- [4] Benjamin Billot, Colin Magdamo, You Cheng, Steven E. Arnold, Sudeshna Das, and Juan Eugenio Iglesias. Robust machine learning segmentation for large-scale analysis of heterogeneous clinical brain MRI datasets. *Proceedings of the National Academy of Sciences*, 120(9):e2216399120, 2023. 3
- [5] Federico Bolelli, Luca Lumetti, Vittorio Pipoli, Kevin Marchesini, Elisa Ficarra, and Costantino Grana. Tooth-Fairy2: Multi-structure segmentation in CBCT volumes, 2024. MICCAI challenge dataset, University of Modena and Reggio Emilia. 6, 2
- [6] Vicent Caselles-Ballester, Eloy Martínez-Heras, Giuseppe Pontillo, Zoe Mendelsohn, Elena M. Marrón, Juan Luis García Fernández, Laia Subirats, Jon Stutters, Jeremy Chataway, Frederik Barkhof, Sara Llufrui, and Ferran Prados. TimeLesSeg: Unified contrast-agnostic cross-sectional and longitudinal MS lesion segmentation via a stochastic generative model. *arXiv preprint arXiv:2605.07955*, 2026. 3
- [7] Stefano Cerri, Oula Puonti, Dominik S. Meier, Jens Wuerfel, Mark Muehlau, Hartwig R. Siebner, and Koen Van Leemput. A contrast-adaptive method for simultaneous whole-brain and lesion segmentation in multiple sclerosis. *NeuroImage*, 225:117471, 2021. 3
- [8] Boqi Chen, Yuanzhi Zhu, Yunke Ao, Sebastiano Caprara, Reto Sutter, Gunnar Rätsch, Ender Konukoglu, and Anna Susmelj. Generalizable single-source cross-modality medical image segmentation via invariant causal mechanisms. *arXiv preprint arXiv:2411.05223*, 2024. 3
- [9] Tugba Akinci D’Antonoli, Lucas K. Berger, Ashraya K. Indrakanti, et al. TotalSegmentator MRI: Robust sequence-independent segmentation of multiple anatomic structures in MRI. *Radiology*, 314(2), 2025. 6, 2
- [10] Maria Correia de Verdier, Rachit Saluja, Louis Gagnon, Dominic LaBella, Ujjwal Baid, Nourel Hoda Tahon, Martha Foltyn-Dumitru, Jikai Zhang, Ahmed W. Moawad, et al. The 2024 brain tumor segmentation (BraTS) challenge: Glioma segmentation on post-treatment MRI. *arXiv preprint arXiv:2405.18368*, 2024. 1
- [11] Michael D. Ernst. Permutation methods: A basis for exact inference. *Statistical Science*, 19(4):676–685, 2004. 3
- [12] Bruce Fischl. FreeSurfer. *NeuroImage*, 62(2):774–781, 2012. 6, 1
- [13] Lidia Garrucho, Kaisar Kushibar, Claire-Anne Reidel, Smriti Joshi, Richard Osuala, Kristina Lang, Oliver Diaz, Apolline Gonzalez-Huete, and Karim Lekadir. MAMA-MIA: A large-scale multi-center breast cancer DCE-MRI benchmark dataset with expert segmentations, 2025. Distributes the Duke-Breast-Cancer-MRI cohort (Synapse syn60868042). 6, 2
- [14] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020. 6
- [15] Qian Guan, Yuqing Chen, Zhun Wei, Ali Asghar Heidari, Huiling Hu, Xiao-Hong Yang, Jiao Zheng, Qiang Zhou, Huiling Chen, and Feng Chen. Medical image augmentation for lesion detection using a texture-constrained multichannel progressive GAN. *Computers in Biology and Medicine*, 145:105444, 2022. progressive GAN augmentation with an explicit texture-discrimination loss, added because standard GAN augmentation lost textural fidelity and hurt lesion detection. 3
- [16] Eldad Haber and Jan Modersitzki. Intensity gradient based registration and fusion of multi-modal images. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 726–733, 2006. 8
- [17] Alexander F. Haddad, Jacob S. Young, Ramin A. Morshed, and Mitchel S. Berger. FLAIRectomy: Resecting beyond the contrast margin for glioblastoma. *Brain Sciences*, 12(5):544, 2022. histopathological tumour infiltration inside peritumoral FLAIR abnormality, beyond the enhancing margin. 2, 10
- [18] Robert M. Haralick, K. Shanmugam, and Its’hak Dinstein. Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-3(6):610–621, 1973. foundational GLCM paper: second-order gray-level co-occurrence statistics carry discriminative content beyond intensity. 3
- [19] Tobias Heimann, Bram Van Ginneken, Martin A Styner, Yulia Arzhaeva, Volker Aurich, Christian Bauer, Andreas Beck, Christoph Becker, Reinhard Beichel, György Bekes, et al. Comparison and evaluation of methods for liver segmentation from CT datasets. *IEEE Transactions on Medical Imaging*, 28(8):1251–1265, 2009. 6, 1
- [20] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. 6, 3
- [21] Nola M. Hylton et al. Locally advanced breast cancer: MR imaging for prediction of response to neoadjuvant chemotherapy—results from ACRIN 6657/I-SPY TRIAL. *Radiology*, 263(3):663–672, 2012. 6, 2
- [22] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. 2, 3, 5, 1
- [23] Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, Zhen Li, Lingyan Zhanng, Wanwan Ma, Xiang Wan, et al. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in Neural Information Processing Systems*, pages 36722–36732, 2022. 6, 1
- [24] Myeongkyun Kang, Philip Chikontwe, Dongkyu Won, Miguel Luna, and Sang Hyun Park. Structure-preserving image translation for multi-source medical image domain adaptation. *Pattern Recognition*, 144:109840, 2023. explicit

- structure-vs-texture disentanglement in cross-domain translation via a texture co-occurrence loss; motivated by naive translation distorting fine texture and degrading downstream segmentation – closest prior precedent for texture preservation as a design principle. 3
- [25] A. Emre Kavur, N. Sinem Gezer, Mustafa Barış, Sinem Aslan, Pierre-Henri Conze, Vladimir Groza, Duc Duy Pham, Soumick Chatterjee, Philipp Ernst, Savaş Özkan, et al. CHAOS challenge – combined (CT-MR) healthy abdominal organ segmentation. *Medical Image Analysis*, 69:101950, 2021. 6, 10, 1
- [26] Frithjof Kruggel, Jasmine S. Paul, and Hermann-Josef Gertz. Texture-based segmentation of diffuse lesions of the brain’s white matter. *NeuroImage*, 39(3):987–996, 2008. motivates texture-based over boundary/edge-based segmentation because diffuse WM lesions are inhomogeneous, unsharp and faint – implicit precedent for the texture-defined vs boundary-defined distinction. 3
- [27] Philippe Lambin, Emmanuel Rios-Velazquez, Ralph Leijenaar, Sara Carvalho, Ruben G. P. J. van Stiphout, Patrick Granton, Catharina M. L. Zegers, Robert Gillies, Ronald Boellard, Andre Dekker, and Hugo J. W. L. Aerts. Radiomics: Extracting more information from medical images using advanced feature analysis. *European Journal of Cancer*, 48(4):441–446, 2012. founding radiomics position paper; quantitative texture features carry prognostic signal across CT/MRI/PET and cancer types. 3
- [28] Pablo Laso, Stefano Cerri, Annabel Sorby-Adams, Jennifer Guo, Farrah Mateen, Philipp Goebel, Jiaming Wu, Peirong Liu, Hongwei Li, Sean I. Young, Benjamin Billot, Oula Puonti, et al. Quantifying white matter hyperintensity and brain volumes in heterogeneous clinical and low-field portable MRI. *arXiv preprint arXiv:2312.05119*, 2023. 3
- [29] Žiga Lesjak, Alfiia Galimzianova, Aleš Koren, Matjaž Lukač, Koen Van Leemput, and Žiga Špiclin. A novel public MR image dataset of multiple sclerosis patients with lesion segmentations based on multi-rater consensus. *Neuroinformatics*, 16(1):51–63, 2018. 1
- [30] Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy M. Hospedales. Learning to generalize: Meta-learning for domain generalization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. 3
- [31] Heng Li, Haojin Li, Jianyu Chen, Mingyang Ou, Hai Shu, and Heng Miao. RaffeSDG: Random frequency filtering enabled single-source domain generalization for medical image segmentation. *arXiv preprint arXiv:2405.01228*, 2024. 3
- [32] Wen Li, David C. Newitt, Jessica Gibbs, Lisa J. Wilmes, Ella F. Jones, Vignesh A. Arasu, and Nola M. Hylton. I-SPY 2 breast dynamic contrast enhanced MRI trial (ISPY2), 2022. Version 1, data set. 6, 2
- [33] Quande Liu, Cheng Chen, Jing Qin, Qi Dou, and Pheng-Ann Heng. FedDG: Federated domain generalization on medical image segmentation via episodic learning in continuous frequency space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 1, 3
- [34] Nathan Molinier, Hendrik Möller, Thomas Dagonneau, Anna Curto-Vilalta, Robert Graf, Matan Atad, Daniel Rueckert, Jan S. Kirschke, and Julien Cohen-Adad. One sequence to segment them all: Efficient data augmentation for CT and MRI cross-domain 3D spine segmentation. *arXiv preprint arXiv:2605.03098*, 2026. 1, 2, 3, 4, 5, 6
- [35] Ju-Hyeon Nam, Nur Suriza Syazwany, Su Jung Kim, and Sang-Chul Lee. Modality-agnostic domain generalizable medical image segmentation by multi-frequency in multi-scale attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3
- [36] Cheng Ouyang, Chen Chen, Surui Li, Zeju Li, Chen Qin, Wenjia Bai, and Daniel Rueckert. Causality-inspired single-source domain generalization for medical image segmentation. *IEEE Trans. Med. Imaging*, 42(4):1095–1106, 2023. 1, 3
- [37] Cheng Ouyang, Chen Chen, Surui Li, Zeju Li, Chen Qin, Wenjia Bai, and Daniel Rueckert. Causality-inspired single-source domain generalization for medical image segmentation. *IEEE Transactions on Medical Imaging*, 42(4):1095–1106, 2023. validated on cross-modality (CT-MRI) abdominal and cross-sequence cardiac MRI segmentation; argues domain-specific spurious cues break on unseen domains, directly relevant to our contrast-specific-cue account. 6
- [38] Savannah C. Partridge et al. Diffusion-weighted MRI findings predict pathologic response in neoadjuvant treatment of breast cancer: The ACRIN 6698 multicenter trial. *Radiology*, 289(3):618–627, 2018. 6, 2
- [39] Gasper Podobnik, Primož Strojani, Primož Peterlin, Bulat Ibragimov, and Tomaz Vrtovec. HaN-Seg: The head and neck organ-at-risk CT and MR segmentation dataset. *Medical Physics*, 50(3):1917–1927, 2023. 6, 2
- [40] Qiang Qiao, Wenyu Wang, Meixia Qu, Kun Su, Bin Jiang, and Qiang Guo. Medical image segmentation via single-source domain generalization with random amplitude spectrum synthesis. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. 3
- [41] Patrik F. Raudaschl et al. Evaluation of segmentation methods on head and neck CT: Auto-segmentation challenge 2015. *Medical Physics*, 44(5):2020–2036, 2017. 6, 2
- [42] Ashirbani Saha, Michael R. Harowicz, Lars J. Grimm, Connie E. Kim, Sujata V. Ghatge, Ruth Walsh, and Maciej A. Mazurowski. A machine learning approach to radiogenomics of breast cancer: a study of 922 subjects and 529 DCE-MRI features. *British Journal of Cancer*, 119(4):508–516, 2018. 6, 2
- [43] Sourya Sengupta, Satrajit Chakrabarty, Keerthi Sravan Ravi, Gopal Avinash, and Ravi Soni. SynthFM: Training modality-agnostic foundation models for medical image segmentation without real medical data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3
- [44] Harshay Shah, Kaustav Tamuly, Aditi Raghunathan, Prateek Jain, and Praneeth Netrapalli. The pitfalls of simplicity bias in neural networks. In *Advances in Neural Information Processing Systems*, 2020. 7
- [45] Hyungseob Shin, Hyeongyu Kim, Sewon Kim, Yohan Jun, Taejoon Eo, and Dosik Hwang. SDC-UDA: Volumetric unsupervised domain adaptation framework for slice-direction

- continuous cross-modality medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2
- [46] Zixian Su, Kai Yao, Xi Yang, Qiufeng Wang, Jie Sun, and Kaizhu Huang. Rethinking data augmentation for single-source domain generalization in medical image segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023. 1, 3
- [47] Kenji Suzuki, Hieu Trung Huynh, Yipeng Liu, Dominic Calabrese, Karen Zhou, Aytekin Oto, and Masatoshi Hori. Computerized segmentation of liver in hepatic CT and MRI by means of level-set geodesic active contouring. *Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2013:2984–2987, 2013. PMC4283827; edge-driven level-set/geodesic-active-contour segmentation of liver – cited as evidence that a gradient ridge exists at organ boundaries. Original entry mis-cited a different (2007, non-matching-title) author list under this PMC id; corrected 2026-08-03 after verification. 10
- [48] Jun Tang, Liangpeng Chen, Ziyang Li, Cong Li, and Deling Li. Imaging strategies for delineating glioma margins: Aiming at precise therapy. *Fundamental Research*, 6(4):2644–2653, 2025. "gliomas do not exhibit a clear boundary between malignant and healthy tissue"; infiltration several cm into normal-appearing parenchyma. Journal corrected 2026-08-03 (was mis-attributed to "Meta-Radiology", a different, unrelated journal). 2, 10
- [49] Franz Thaler, Martin Urschler, Mateusz Kozinski, Matthias A. F. Gsell, Gernot Plank, and Darko Stern. Semantic-aware random convolution and source matching for domain generalization in medical image segmentation. *IEEE Access*, 2025. 2, 3
- [50] Romain Valabregue, Ines Khemir, Eric Badinet, François Rousseau, Guillaume Auzias, and Reuben Dorent. SIAM: Head and brain MRI segmentation from few high-quality templates via synthetic training. *arXiv preprint arXiv:2605.02737*, 2026. 3
- [51] Jasper W. van der Graaf, Miranda L. van Hooff, Constantinus F. M. Buckens, Matthieu Rutten, Job L. C. van Susante, Robert Jan Kroeze, Marinus de Kleuver, Bram van Ginneken, and Nikolas Lessmann. Lumbar spine segmentation in MR images: a dataset and a public benchmark. *Scientific Data*, 11:264, 2024. 6, 2
- [52] Shujun Wang, Lequan Yu, Kang Li, Xin Yang, Chi-Wing Fu, and Pheng-Ann Heng. DoFE: Domain-oriented feature embedding for generalizable fundus image segmentation on unseen datasets. *IEEE Trans. Med. Imaging*, 39(12):4237–4248, 2020. 1, 3
- [53] Shaun Warrington, Andrea Torchi, Olivier Mougin, Jon Campbell, Asante Ntata, Martin Craig, Stephanos Assimpopoulos, Fidel Alfaro-Almagro, Karla L. Miller, Mark Jenkinson, Paul S. Morgan, and Stamatios N. Sotiropoulos. A multi-site, multi-modal travelling-heads resource for brain MRI harmonisation. *Scientific Data*, 12:595, 2025. 6, 1
- [54] Jakob Wasserthal et al. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023. 6, 2
- [55] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. 3
- [56] Zhenlin Xu, Deyi Liu, Junlin Yang, Colin Raffel, and Marc Niethammer. Robust and generalizable visual representation learning via random convolutions. In *International Conference on Learning Representations (ICLR)*, 2021. 1, 3
- [57] Jing Zhang, Longzheng Tong, Lei Wang, and Ning Li. Texture analysis of multiple sclerosis: a comparative study. *Magnetic Resonance Imaging*, 26(8):1160–1166, 2008. PMID 18513908; texture features discriminate MS lesion vs normal-appearing WM vs normal WM. 2, 10
- [58] Ling Zhang, Xiaosong Wang, Dong Yang, Thomas Sanford, Stephanie Harmon, Baris Turkbey, Bradford J. Wood, Holger Roth, Andriy Myronenko, Daguang Xu, and Ziyue Xu. Generalizing deep learning for medical image segmentation to unseen domains via deep stacked transformations. *IEEE Trans. Med. Imaging*, 39(7):2531–2540, 2020. 1, 3
- [59] Haoyu Zhao, Wenhui Dong, Rui Yu, Zhou Zhao, Du Bo, and Yongchao Xu. MoreStyle: Relax low-frequency constraint of fourier-based image reconstruction in generalizable medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. 3

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

7. Training configuration

All methods share one nnU-Net [22] configuration; only the input augmentation differs. Nothing below is tuned per dataset.

Training schedule. 3D full-resolution nnU-Net, SGD with Nesterov momentum 0.99, initial learning rate 10^{-2} with the nnU-Net polyLR decay, 250 mini-batches per epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the nnU-Net planner selects for that dataset, identically for every method. The epoch budget is fixed per dataset and shared by all methods on it: 2500 (Brats-GLI), 2000 (ON-Harmony), 2000 (Open-MS), 200 (CHAOS). Every number reported in this paper is a 3-fold cross-validated average; folds are the same across methods.

PALETTE hyper-parameters. Set once and never tuned per dataset: k -means class count drawn per sample from $\{2, \dots, 6\}$, skipped in favour of a single global remap with probability 0.10; Voronoi sub-parcellation of each intensity class with probability 0.60 (else left as one region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-region as in Eq. (1) (always applied, to whatever partition results from the k -means/Voronoi steps); per-label affine decoupling applied independently per label with probability 0.5 (skipped for any label with under 4 foreground voxels in that sample); Gaussian blur σ drawn from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level and after the label-level remap), so mostly not applied at either point. The transform is applied on the GPU, on-the-fly, to a fixed 50% of training iterations (*train050*); the remaining half of iterations use the un-transformed real image.

Validation-time configuration. *val000* (the configuration reported in the main paper) applies no synthetic contrast to the validation split, matching every competing method and keeping model selection like-for-like. *val100* applies PALETTE to the entire validation split; it is reported in Sec. 10.

Baseline configurations. SynthSeg-noEM uses the authors' released generator with its default priors. SynthSeg-EM is our re-implementation of the variant described but not released in the original work; it estimates a per-class intensity model from the sparse task labels available in

each dataset via expectation-maximization, and is otherwise identical in its sampling procedure. SRCSM and Auglab use their published default configurations. No competing method was re-tuned in our favour, and all use the same folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs. A single fold costs from under an hour (CHAOS, 200 epochs) to roughly three days (Brats-GLI, 2500 epochs); the full study is dominated by the 7 methods $\times$ 8 settings $\times$ 3 folds main comparison plus the ablation ladders of Sec. 4.3.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. For each task the model is trained on a single source contrast and evaluated on every contrast of that task's test split, including the training contrast itself; training and validation subjects never appear in any reported number.

ON-Harmony [53] — 31-region healthy-brain segmentation. Reference labels are *silver standard*: they were produced automatically with FreeSurfer [12] on the T1w volume of each subject and then registered to that subject's other contrasts, which are acquired in the same session. They are therefore not manual annotations, and absolute Dice on this task should be read as agreement with an automated reference rather than with a human rater; the comparison between methods, which all use the identical reference, is unaffected.

Open-MS [29] — MS-lesion segmentation with manual consensus annotations, evaluated across FLAIR, T1w and T2w.

Brats2024-GLI [10] — post-treatment glioma sub-region segmentation, evaluated across T1n, T1c, T2w and T2-FLAIR.

CHAOS [25] — abdominal organ segmentation (liver, spleen, both kidneys), evaluated across the CHAOS MR contrasts and, as a cross-modality transfer, on held-out CT from AMOS [23] and SLIVER07 [19].

ToothFairy2, HaN-Seg and PDDCA [5, 39, 41] — mandible segmentation, trained on dental cone-beam CT, the one non-CT, non-MRI training modality here. ToothFairy2’s annotation coverage is cohort-dependent, so every column of this task is scored on the mandible alone (teeth excluded, matching the external references), which keeps the numbers being averaged commensurable. ToothFairy2 acquires CBCT only, so its held-out evidence is entirely cross-dataset: HaN-Seg head-and-neck CT and MR-T1 and the PDDCA CT, the two CT sources pooled as one contrast group. ToothFairy2’s volumes are stored superior-inferior inverted relative to the external cohorts — an inversion invisible to the image header, found by checking anatomy directly — so the external test volumes are flipped into ToothFairy2’s convention before inference, which leaves Dice and HD95 unchanged under the rigid transform. The correction matters for the comparison: a method without flip augmentation (SRCSM) was penalised by the inversion far more than the others, and it is after this correction that SRCSM leads this task.

Breast: I-SPY2, Duke-Breast-MRI, I-SPY1, ACRIN-6698 [13, 21, 32, 38, 42] — breast-lesion segmentation on malignant-only cohorts. I-SPY2 (CC BY 4.0) trains, on its first-post-contrast T1 (T1-CE) and its T2w; its reference is the trial’s functional tumour volume mask. Three external cohorts are test-only. Duke-Breast-MRI and I-SPY1 come through the MAMA-MIA release, whose expert primary-lesion protocol both share, and contribute T1-CE and pre-contrast columns; Duke-Breast-MRI is CC BY-NC 4.0, a non-commercial term no other cohort here carries. ACRIN-6698, the diffusion sub-study of I-SPY2 with no patient in common with our I-SPY2 subjects, contributes the baseline diffusion-weighted acquisition with the trial’s whole-tumour DWI region, a contrast neither training modality resembles. Because I-SPY2 is mostly acquired unilaterally, bilateral external volumes are cropped to the lesion side and, for the full-thorax acquisitions, also anterior-posteriorly to the breast-plus-chest-wall extent of I-SPY2’s own crops; without the second crop every method’s HD95 is dominated by the thorax. I-SPY2’s DCE series are acquired unilaterally at some sites and bilaterally at others; both field-of-view variants are kept per patient, and the significance tests treat them as one patient.

Spine [51] — vertebrae, intervertebral discs and spinal canal. Results for this task were computed by our collaborators on their own pipeline with the same method suite, trained on CT and Dixon in-phase MRI of a CT/Dixon cohort and tested on that cohort’s held-out CT and Dixon fat, in-phase and water images and on the T1w and T2w of SPIDER. Their Dice is a whole-structure binary Dice per label; no HD95 and no SRCSM arm were computed, so Spine is

absent from the HD95 table and from every SRCSM comparison.

Pelvis [9, 54] — ten pelvic and hip musculoskeletal structures (both hips, sacrum, and the gluteus maximus, medius and minimus and iliopsoas on each side), from the TotalSegmentator CT (CC BY 4.0) and MRI (CC BY-NC-SA 2.0) releases. The two are unpaired cohorts, and the MRI release mixes sequences by design, so its MRI column is a sequence-heterogeneous test rather than one named contrast. CT is subsampled to 200 cases to keep training time comparable with the other tasks.

Task	Train	Test
CHAOS	16	24
ON-Harmony	84	8
Open-MS	22	8
Brats2024-GLI	630	70
ToothFairy2	408	71
I-SPY2	476	84
Pelvis (CT / MRI)	170 / 156	30 / 27

Table 5. Per-task subject counts for the training cohorts, pooled across the 3 folds used throughout (± 1 between folds). ON-Harmony counts are scan sessions; the eight test sessions are four subjects, each scanned at two sites. External cohorts are test-only and counted separately: AMOS and SLIVER07 (CHAOS), HaN-Seg and PDDCA (Mandible), Duke-Breast-MRI (291), I-SPY1 (167) and ACRIN-6698 (371) for Breast, and SPIDER (171 T1w, 219 T2w) for Spine, whose own CT/Dixon test split has 27–29 cases per contrast. I-SPY2 counts are patients; one held-out patient can yield more than one T2w image (both field-of-view variants). Pelvis MRI keeps 183 of the 298 released cases.

9. Per-setting results

Tab. 2 in the main paper reports results at the task level, pooling each task’s held-out test contrasts, all of its source modalities and its external cohorts. This section reports the same evaluation broken out into the eighteen individual source-modality settings. Every number is a cross-fold, cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Statistical test construction. All tests are paired, and the unit of independence is the *patient*, never the fold, the image or the contrast. A patient’s score is its mean over labels and evaluated folds; where the same patient is scored in several held-out contrasts, or by the models of several training modalities, those scores move together as one unit under the null. Treating them as separate samples would be pseudo-replication and would inflate the effective sample size — a real risk here, since several tasks test a handful of patients

under many contrasts (ON-Harmony: eight sessions of four subjects, six contrasts). For ON-Harmony the unit is the session, since its four subjects leave a test that could never reach significance; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic, and we do not rest any claim on ON-Harmony alone. Per-setting comparisons (below) use the paired Wilcoxon signed-rank test [55], since there each setting is judged on its own. All tests are Holm-corrected [20] for multiple comparisons.

Why the aggregate tests are sign-flip tests on a macro-average. The aggregate tests (Tab. 2 and the per-dataset headline blocks) are not paired Wilcoxon signed-rank tests, which is the more usual choice in this literature: SynthSeg+, from the lineage we compare against, marks a method as better than its competitors using a one-sided Bonferroni-corrected Wilcoxon signed-rank test on paired Dice [4]. We therefore give our reason rather than leave it implicit. The *estimand* we test, by contrast, is not unusual at all — it is the same equal-weight-per-stratum principle the Medical Segmentation Decathlon adopted, for the same reason, as set out below.

The estimand. What we claim is $\text{macro}\Delta$: the mean, over held-out contrasts, of that contrast’s mean paired per-case difference (ours minus competitor) — equal weight per contrast, which is exactly what the *all* column of every table reports, so the test and the table measure the same thing. Equal weighting is not cosmetic here. The claim under test is cross-contrast generalization, and a contrast’s held-out case count reflects how large a public cohort happens to be, not how much that contrast should say about generalizing. CHAOS is the extreme case: its held-out CT evidence includes an external cohort contributing 200 cases, while each of its three MR contrasts contributes 8. Weighting by case count would let one CT cohort cast more than half the votes in a test whose conclusion is about transfer across MR contrasts.

This is the same problem, and the same answer, as the Medical Segmentation Decathlon’s. Facing sample sizes that “varied heavily across all tasks and target ROIs,” it ranked algorithms within each target ROI, averaged those ranks per task, and averaged the per-task ranks for the final score — choosing that over a flat average across ROIs precisely because the flat scheme means “tasks with multiple target ROIs ... would be over-weighted” [1]. Our $\text{macro}\Delta$ applies the same principle one level across, with the held-out contrast in place of the target ROI. Where we differ is only in how that weighted quantity is reached: the Decathlon arrives at it through counts of pairwise Wilcoxon outcomes at an uncorrected per-comparison α , whereas we test the weighted difference itself and Holm-correct across competitors.

Why not a pooled Wilcoxon. The signed-rank test is

defined on a single undifferentiated sample of paired differences. Applying it to the concatenation of all contrasts therefore tests a case-weighted null — the imbalance above, reintroduced — and there is no standard reweighting of the signed-rank statistic that recovers the equal-per-contrast estimand. The disagreement this causes runs in both directions: a pooled test can declare a tenth of a Dice point significant because one large cohort supplies most of the cases, or miss a consistent effect confined to the smaller contrasts.

The test. Under the null that the two methods are exchangeable on a given patient, that patient’s paired differences are equally likely to carry either sign. Sign-flipping is therefore a valid randomization null for *any* statistic computed from those differences [11], $\text{macro}\Delta$ included, which is what makes it applicable where the signed-rank test is not. We report the one-sided p in the direction of the stated hypothesis, with the two-sided p retained as a secondary column, and give a 95% confidence interval from a bootstrap that resamples patients within groups of patients sharing the same set of contrasts, so each source keeps its observed size.

Evaluating the tail. The statistic is a weighted sum of per-patient contributions, so its sign-flip null distribution can be computed rather than sampled. We enumerate all 2^M sign patterns of the $M=20$ patients with the largest contributions exactly, and treat the remaining, lighter patients as a Gaussian term with their exact null variance; with twenty patients or fewer the test is therefore exact. A plain normal approximation to the whole null, which an earlier version of this analysis used, is badly conservative in exactly the designs we have, where a few patients spanning several contrasts carry most of the null variance: on the CHAOS table it returned $p = 0.013$ for a comparison that a 200,000-draw Monte-Carlo test places below 5×10^{-6} . The enumerated tail is deterministic and reaches far tails without the $1/B$ floor of resampling. Its corollary is a hard floor for small cohorts: eight patients admit $2^8 = 256$ sign patterns, so no one-sided p can fall below $1/256$ however consistent the effect.

What is reported alongside. Every generated significance report also carries a per-contrast table of mean Δ with the raw, uncorrected Wilcoxon p for each contrast individually, and the per-setting tables in this section are Wilcoxon throughout. The standard test is reported; it is simply not the one the pooled headline number rests on.

Contrast-group pooling. For CHAOS, the same tested modality is sometimes evidenced by more than one source: CHAOS’s CT evidence comes from its own held-out CT split plus AMOS and SLIVER07. Averaging every raw column with equal weight — the natural first implementation — silently lets a modality with more contributing sources outvote one with fewer in both the displayed *all* average and the significance test, since it then supplies more of

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	25.3*	8.7*	43.1*	40.3*	46.3	46.4
Brats-GLI T1c	28.3*	9.5*	42.9*	44.5*	46.6	47.1
Brats-GLI T2w	18.1*	8.4*	44.1*	43.8*	47.1	47.3
Brats-GLI FLAIR	21.8*	8.7*	42.3	37.5*	39.4*	43.0
CHAOS T1in	29.4*	62.6*	83.4*	83.7	83.1*	85.1
CHAOS T2spir	29.1*	56.6*	82.7*	84.1*	83.5*	84.8
ON-Harmony T1w	27.5*	66.0*	67.3*	66.4	67.6*	68.6
ON-Harmony T2w	34.8*	66.7	67.5	66.7	67.2	67.7
ON-Harmony DWI	42.6*	61.6	64.2	66.8	63.4	63.1
Open-MS FLAIR	25.2*	4.6*	32.8*	34.6*	32.8*	39.5
Open-MS T1w	16.7*	4.0*	35.1*	31.4*	35.8	37.4
Mandible CBCT	59.2*	53.6*	76.3*	79.0	78.3	77.7
Breast T1-CE	34.1*	16.6*	40.4*	18.0*	41.1*	41.9
Breast T2w	35.9*	16.5*	38.9*	14.8*	39.0*	40.5
Spine CT	27.0*	49.3*	79.1*	—	77.5*	80.2
Spine Dixon in	69.9*	51.6*	82.2*	—	84.2	84.3
Pelvis CT	35.6*	17.1*	66.5*	71.4	68.5	69.2
Pelvis MRI	24.0*	12.8*	58.6*	59.2*	60.0	62.0

Table 6. Per-setting Dice (%), $\uparrow$, best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 2, so the rows of a task average to that column. *: PALETTE significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run and no HD95.

the equally-weighted columns being averaged. We instead pool disjoint sources of the *same* modality by concatenating their cases into one group before any averaging happens, then average modality groups with equal weight – for CHAOS this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS’s own CT column already blends all three organs into one number before it reaches this step. The same pooling applies to Mandible (HaN-Seg and PDDCA both supply CT) and Breast (I-SPY2, Duke-Breast-MRI and I-SPY1 all supply T1-CE and pre-contrast); the remaining tasks have no contrast evidenced by more than one source.

Tables 6 and 7 break each task column of Tab. 2 into its training modalities. Where a margin is not significant in a single setting, the honest reading is that one setting cannot separate the methods, not that they are equivalent: several settings have few test units (eight patients on Open-MS, eight sessions on ON-Harmony), which is why the main claims rest on the task- and cross-task-level tests.

Boundary distance at task level. Tab. 8 gives the HD95 counterpart of Tab. 2. Across tasks PALETTE is significantly better than every competitor, Auglab included. Per task it has the lowest HD95 on Brats-GLI, ON-Harmony, Open-MS and Pelvis; it ties Auglab on Breast and trails SRCSM on CHAOS without either gap being significant; and on Mandible it is significantly behind SRCSM, Auglab

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	41.0*	78.0*	18.4*	20.8*	15.1	15.4
Brats-GLI T1c	40.3*	78.3*	19.5*	14.9	15.1	15.4
Brats-GLI T2w	54.9*	79.7*	16.5*	15.7*	14.8	15.0
Brats-GLI FLAIR	43.1*	76.0*	17.3	18.5*	19.2*	16.6
CHAOS T1in	120.0*	87.5*	33.8	26.3	43.5*	30.3
CHAOS T2spir	128.8*	92.4*	29.8	27.2	35.1*	30.9
ON-Harmony T1w	33.4*	10.4*	5.4	6.7	5.2*	4.3
ON-Harmony T2w	23.7*	4.6*	4.0	4.6*	4.2*	4.0
ON-Harmony DWI	14.7*	4.9	4.6	4.5	4.6	4.7
Open-MS FLAIR	39.9*	41.5*	23.6	23.8	26.6	24.3
Open-MS T1w	30.5*	42.3*	25.9	25.5	26.3*	24.2
Mandible CBCT	21.7	32.9*	24.9*	19.3	22.1	23.5
Breast T1-CE	61.8*	103.2*	61.7*	92.4*	57.3	59.0
Breast T2w	55.2	99.1*	60.1*	91.2*	56.9	56.3
Pelvis CT	146.2*	252.0*	20.0	16.0	23.5	21.6
Pelvis MRI	152.3*	271.6*	48.0	51.9*	47.8*	36.4

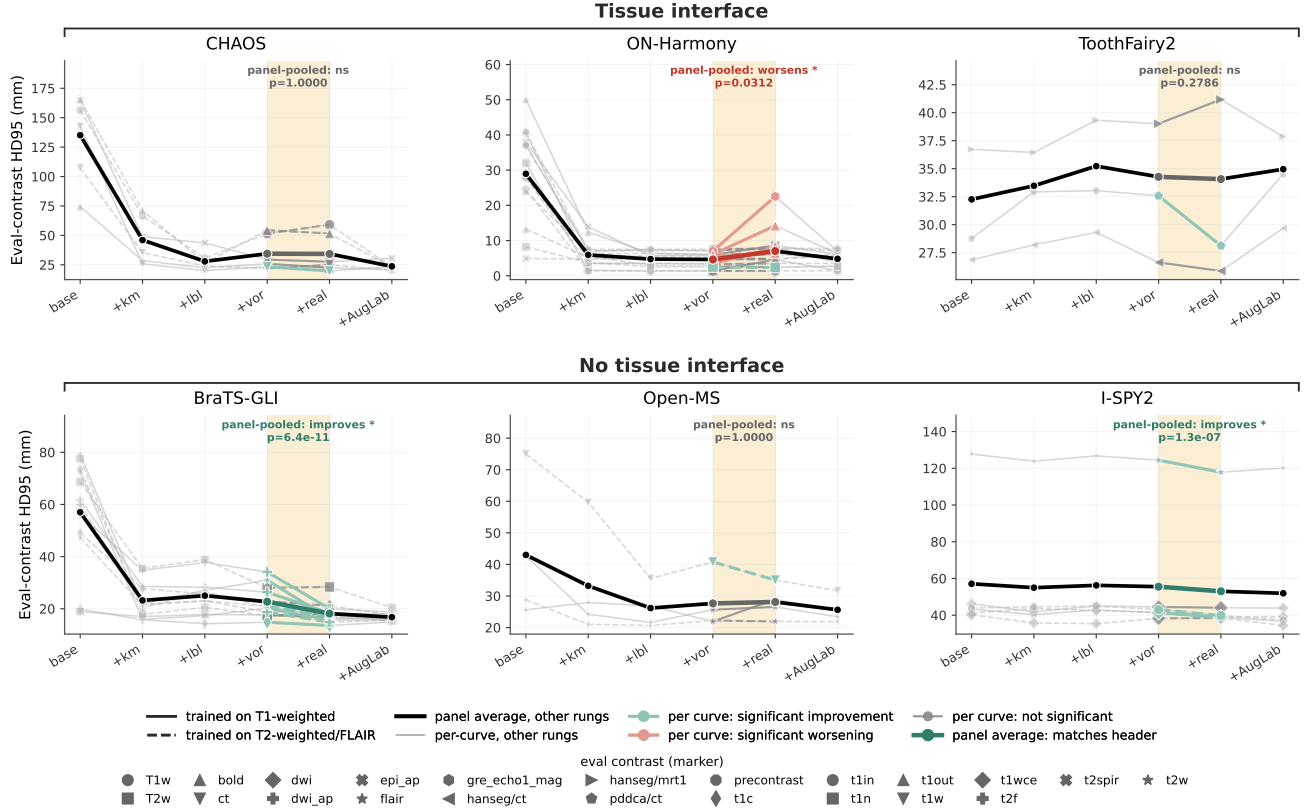
Table 7. Per-setting HD95 (mm, $\downarrow$), best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 2, so the rows of a task average to that column. *: PALETTE significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run and no HD95.

and the no-augmentation baseline, the one task where the baseline beats anything. When HD95 disagrees with Dice for a method – a noticeably better HD95 than its Dice would predict – the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the structure can have a deceptively good boundary distance while missing most of the target, so a strong HD95 paired with a weak Dice is a warning sign rather than evidence of competitiveness.

Full ladder: numbers and significance. Fig. 2 (main paper) shows the rung sequence visually; Tab. 10 gives the underlying out-of-domain Dice at every rung of all fourteen ladders, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon on patients, within each ladder’s family of steps). Most steps that add an ingredient help, as expected. Six cells are significant *decreases* (marked $\downarrow$). Two are the fill-swap step on ON-Harmony T1w and T2w, consistent with its tissue-interface grouping. One is the Voronoi sub-parcellation step on Brats-GLI T1n, which the real-fill step then more than recovers. The other three are the final +Auglab rung — on ON-Harmony DWI and twice on Mandible, the task where the complete method places third (Sec. 4.2). There PALETTE alone (the +real rung, 69.5) is the ladder’s maximum and adding the Auglab recipe takes it back down, which places the Mandible shortfall in the wrapper rather than in the transform this paper argues for. We have not diagnosed which Auglab operator is responsible.

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Pelvis	p vs ours
Baseline	44.8	124.4	24.0	35.2	21.7	58.5	149.3	8.4e-33
SynthSeg-noEM	78.0	89.9	6.7	41.9	32.9	101.1	261.8	2.1e-39
SynthSeg-EM	17.9	31.8	4.7	24.7	24.9	60.9	34.0	4.7e-6
SRCSM	17.5	26.7	5.3	24.6	19.3	91.8	33.9	3.0e-23
Auglab	16.1	39.3	4.7	26.5	22.1	57.1	35.6	5.1e-6
PALETTE (ours)	15.6	30.6	4.3	24.3	23.5	57.7	29.0	—

Table 8. Task-level HD95 (mm, ↓). Same pooling and test as Tab. 2, and likewise no cross-task average. Spine reports no HD95.

Figure 4. The causal-ablation ladder of Fig. 2 (main paper), same per-dataset panels, same boundary-type grouping, same fill-swap rung and same visual scheme (per-curve colour only on the real-fill segment; bold black panel average; one panel-pooled p -value per panel), reported on HD95 instead of Dice (*improvement* means the fill-swap step lowers HD95). Brats-GLI and Breast improve significantly at panel level and ON-Harmony worsens; CHAOS, Open-MS and ToothFairy2 do not move significantly. Underlying per-row deltas in Tab. 9.

We likewise do not hide that Voronoi’s own marginal contribution is task-dependent rather than uniformly positive – a separate paired significance check (same protocol as Tab. 14’s all-contrast Dice, not just the OOD slice used above) confirms the same pattern: Voronoi significantly *helps* Open-MS FLAIR and (by a small margin) ON-Harmony T1w, is flat on CHAOS T1in, and significantly *hurts* Brats-GLI T1n under the real-fill deployed configuration as well, not only here under noise fill. The full method still wins on Brats-GLI T1n regardless (Tab. 2): whatever

Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is not.

Per-label breakdown: is the fill-swap effect uniform within a task? Fig. 2 shows that Brats-GLI T2w’s pooled fill-swap effect is small and not significant, with one significantly negative eval contrast inside it (T2w→T1n). This raises a further question: within that one (training, eval)

Task	boundary type	Δ HD95 (mm)	p
CHAOS T1in	interface	-2.95	0.035
CHAOS T2spir	interface	+2.64	0.88
ON-Harmony T1w	interface	+6.82	0.062
ON-Harmony T2w	interface	+0.38	0.062
ON-Harmony DWI	interface	-0.05	0.75
ToothFairy2 CBCT	interface	-0.20	0.37
Open-MS FLAIR	no interface	-3.02	0.14
Open-MS T1w	no interface	+3.94	0.88
Brats-GLI T1n	no interface	-4.59	6.8e-10
Brats-GLI T1c	no interface	-9.91	4.5e-11
Brats-GLI T2w	no interface	-1.95	0.014
Brats-GLI FLAIR	no interface	-1.71	0.21
I-SPY2 T1-CE	no interface	-3.21	9.5e-5
I-SPY2 T2w	no interface	-1.82	0.0026

Table 9. Same fill-swap step as Tab. 3 (main paper), on HD95 (negative is better), with p Holm-corrected across the fourteen rows. Five of the six significant improvements are on no-tissue-interface targets, including both Breast rows, whose Dice does not move; the sixth is CHAOS T1in. HD95 therefore leans the same way as the main paper’s one-directional Dice result, but its sign is less consistent within a task (it flips between the two modalities of Open-MS and of CHAOS), so we state the claim on Dice and report HD95 here rather than as confirmation of it.

Rung	Open-MS FLAIR	Open-MS T1w	Brats-GLI T1n	Brats-GLI T1c	Brats-GLI T2w	Brats-GLI FLAIR	CHAOS T1in
base	2.4	1.1	16.5	15.9	6.0	11.0	21.4
+km	11.2*	9.8*	38.4***	29.6***	38.6***	26.1***	84.2***
+lbl	17.5*	22.4*	41.8***	30.3	40.2**	26.4	87.7***
+vor	19.6	34.1*	39.8***↓	30.5	41.0	31.9***	87.2
+real	27.3*	34.9	47.0***	43.7***	41.8	31.7	88.2***
+AL(v000)	30.6	36.4	46.9	44.2	46.8***	40.9***	87.7
+AL(v100)	31.3	34.7	47.6*	46.5***	46.9	41.2	87.3

Rung	CHAOS T2spir	ON-Harm. T1w	ON-Harm. T2w	ON-Harm. DWI	Mandible CBCT	Breast T1-CE	Breast T2w
base	18.3	14.8	24.5	36.1	39.4	23.7	33.5
+km	79.0***	62.5*	63.5*	60.4*	64.6***	28.8***	37.0***
+lbl	83.4***	62.9	63.6	64.5*	65.6***	28.5	38.8***
+vor	82.7	63.2	64.2*	64.6	68.1***	30.8***	39.8**
+real	81.8	58.0*↓	62.6*↓	65.4*	69.5***	29.9	40.3*
+AL(v000)	86.3	64.2	63.9	60.9*↓	67.6***↓	34.7***	41.9***
+AL(v100)	87.1	64.2	64.4*	60.9	67.5***↓	—	—

Table 10. Out-of-domain Dice (%) at every rung of every ladder (Fig. 2), with significance for each step against the rung above it: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Holm-corrected paired Wilcoxon on patients, within each ladder); ↓ marks a significant *decrease*. Rungs: base(line), + k -means, +label-remap, +Voronoi (noise fill), +real-intensity fill (PALETTE alone), +Auglab (val000/val100). Open-MS and ON-Harmony have eight test units each, so their smallest attainable p after correction is just under 0.05. I-SPY2 has no val100 arm. Mandible’s held-out axis is cross-dataset.

pair, does the fill-swap step affect every anatomical sub-region equally, or is the worsening itself concentrated in specific labels? We break Brats-GLI T2w→T1n down by its four labels – necrotic core (NCR), oedema (SNFH), en-

hancing tumour (ET), resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumour has a necrotic core or a resection cavity).

Label	T1n→T2w	T2w→T1n	gap
NCR	21.5	19.3	+2.1
SNFH	66.3	45.3	+21.0
ET	27.5	27.9	-0.4
RC	52.3	43.1	+9.1

Table 11. Absolute OOD Dice (%) of the deployed real-fill (v26.6.2) model, per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+1.9 (p=0.10)	+0.3 (p=0.87)
SNFH	+14.7 (p=4.3e-11)	-9.8 (p=4.5e-05)
ET	+2.3 (p=0.017)	+0.9 (p=0.38)
RC	+2.9 (p=0.0003)	-6.5 (p=0.0039)

Table 12. Fill-swap step (+voronoi noise fill → v26.6.2 real fill), per label, both directions. p : Holm-corrected paired Wilcoxon, family = the 4 labels within each direction. **Bold** = significant.

The worsening is not a property of “the tumour” – it is concentrated almost entirely in SNFH, which alone accounts for +21.0 of the roughly 14-point whole-tumour Dice gap between directions, and is the only label whose fill-swap delta significantly reverses sign between the two training directions (+14.7 when trained on T1n, -9.8 when trained on T2w). RC shows the same reversal at smaller magnitude. NCR shows no significant effect in either direction, and ET helps a little only in the direction that already works (T1n→T2w), never significantly reversing when trained on T2w. Fig. 5 shows why on one representative case: the T2w-trained model’s real-fill prediction on SNFH collapses to a thin band near the ventricle relative to its own noise-fill prediction on the same case, while its ET prediction looks essentially unchanged between the two rungs.

One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast’s rendering of a diffuse, texture-defined label like oedema may not simply fail to transfer to a different contrast’s rendering of the same tissue – it may actively mislead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an instance of shortcut learning [14] – specifically, a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al. [37] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation – rather than the intended, causal feature, which

BraTSGLI02108101, fold 0 -- T2w-trained model evaluated on T1n

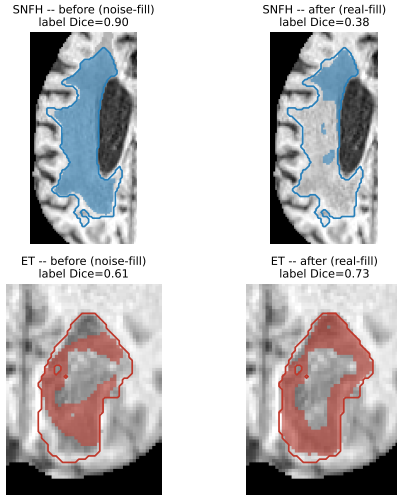


Figure 5. BraTSGLI02108101, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Colour = ground truth wherever the prediction agrees; grey = false positive; outline = ground truth.

a simplicity-biased network need not prefer even when it would generalize better [44]. We have not tested this mechanistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

Validation-time randomization improves most tasks on both metrics (Tab. 13), most visibly on boundary distance for CHAOS (30.6 $\rightarrow$ 28.7 mm). The effect is not significant across tasks, and Open-MS Dice moves slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in

Task	Dice $\uparrow$		HD95 (mm) $\downarrow$	
	val000	val100	val000	val100
Brats-GLI	45.9	46.6	15.6	15.1
CHAOS	85.0	85.2	30.6	28.7
ON-Harmony	66.5	66.6	4.3	4.3
Open-MS	38.5	37.9	24.3	24.2

Table 13. Applying PALETTE at validation time as well as at training time, on the four tasks where both configurations were evaluated (Mandible’s val100 arm was not evaluated on its cross-dataset test sets; the Breast, Spine and Pelvis arms were not trained). Across these tasks the two are statistically indistinguishable in either direction (same cross-task test as Tab. 2; two-sided Dice $p = 0.62$, HD95 $p = 0.18$), though val100 is significantly ahead within Brats-GLI and ON-Harmony individually. We report the like-for-like val000 configuration throughout the main paper.

Component	CHAOS	Open-MS	BraTS	ON-Harmony
Base (no augmentation)	36.1	25.2	25.3	27.5
+ k -means partition	84.8	30.6	40.6	67.6
+ label remap	87.7	34.6	43.2	68.0
+ Voronoi sub-parcellation	87.2	36.1	41.8	68.2
+ affine remap (Ours)	87.9	40.4	47.4	63.6
+ Auglab (deployed)	87.4	39.5	46.4	68.6

Table 14. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE component on top of the row above; all-modality Dice (%), averaged over held-out test contrasts, folds 0–2, train fraction 50%. BraTS and ON-Harmony report their T1n/T1w settings, matching every other table in this paper. The deployed row matches Tab. 6’s **Ours** column exactly.

general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE in its entirety and is denoted **Ours** in this ladder — identical to the standalone (no-Auglab) configuration used throughout Sec. 4.2. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

1483 **Composability with Auglab.** PALETTE is deployed,
1484 throughout this paper, as one additional operator inside
1485 the Auglab pipeline rather than as a replacement for it
1486 (Sec. 4.1); Auglab is a full prior augmentation method in
1487 its own right, not a component of PALETTE, so stacking
1488 it on top of the ladder above is a composability/deployment
1489 check, not a further mechanism-ablation step. Tab. 14 keeps
1490 the two separate: the top five rows isolate PALETTE’s in-
1491 ternal mechanism, evaluated with no Auglab throughout,
1492 while the bottom row reports PALETTE plus Auglab — the
1493 configuration used everywhere else in this paper — set apart
1494 by a rule rather than presented as the next rung of the same
1495 ladder.